Jagannath Saha

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Education

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science — *Jun. 2023 – May 2027*

Aditya Academy Secondary — Kolkata, India
Higher Secondary — *Apr. 2022 – Mar 2023*

Technical Skills

Languages: Java, Python, C/C++, TypeScript, JavaScript, SQL
Frameworks: FastAPI, Spring Boot, React, Tkinter
ML/Data Libraries: Scikit-learn, Librosa, Resemblyzer, NumPy, SciPy
Developer Tools: Git, Docker, Maven, Vite
Core Concepts: Machine Learning, Speaker Recognition, RAG, REST APIs, JWT Authentication, Data Processing, Data Structures, Algorithms

Projects

LearnLite — Offline AI Tutoring Platform | *FastAPI, React, TypeScript, Ollama, Tailwind CSS* | GitHub
- Developed an offline AI tutoring platform using FastAPI, React, and a local LLM (Ollama)
- Implemented prompt engineering with contextual memory and multilingual support
- Enabled real-time streaming via Server-Sent Events (SSE) for responsive interaction
- Built a chat UI with LaTeX rendering and localStorage-based persistence
- Ensured stable inference with validation, retry logic, and request constraints

Nexpenza — Expense Management System | *Java, Spring Boot, React, TypeScript, JWT, JPA* | GitHub
- Built a full-stack expense tracking system for managing and analyzing transactions
- Implemented JWT-based authentication and role-based access control with Spring Security
- Developed REST APIs using JPA with flexible query support
- Created dashboards to visualize spending patterns and insights
- Designed reusable frontend components with structured state management

VoiceAuth — Voice-Based Biometric Authentication | *Python, Tkinter, Librosa, Resemblyzer, Scikit-learn* | GitHub
- Built a voice authentication system using Resemblyzer for speaker embedding-based identity verification
- Implemented MFCC feature extraction and cosine similarity for voiceprint matching
- Integrated SVM (RBF kernel) with similarity scoring to improve classification accuracy
- Developed a Tkinter GUI with real-time waveform and MFCC visualization
- Applied audio preprocessing (noise reduction, normalization, silence trimming) using Librosa

Awards

3rd Place – Data Samurai Competition | *Cintel Student Association, SRMIST* — Feb 2025
- Achieved 3rd place in a data-focused competition involving analytical problem-solving and model development

Experience

NLP Research Intern — May 2025 – July 2025
Jadavpur University — *Kolkata, India*
- Applied Hugging Face Transformers for numerical claim verification to enable automated fact-checking
- Implemented re-ranking techniques to improve relevance and alignment of retrieved evidence
- Leveraged Retrieval-Augmented Generation (RAG) for multi-source reasoning and accurate verification